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1st Periodical Examination (Sep, 2024)

Course Name: - Digital System Design Course Code :-(ECEPC-312) Schemes: NEP

Branch:- CSE (AI&ML) Semester:-3rd Time:-2 Hours

Max Marks: -50

Note:-Attempt all questions.

Course Outcomes:

CO1: To demonstrate understanding of Boolean algebra and binary arithmetic operations.

CO2: To identify and describe MSI and their applications in digital circuits.

CO3: To explain the operation and design considerations for synchronous circuits.

CO4: To interpret and differentiate between dataflow, behavioral, and structural modeling in VHDL.

Q1. (a) Define Implicants and Prime implicants.

(b) State De Morgans Theorem.

(c) Differentiate between Multiplexer and Encoder.

(d) Give classification of floating point numbers representation.

(e) What are the limitations of Parity bit method?

(5*2), CO1

Q2. (a) Minimize the logic function by using K map.

$$F(A,B,C,D) = \sum m(1,3,7,10,11,12,14,15) + d(0,2,5,8)$$

(b) $F(A,B,C,D) = \prod M(1,3,7,12,14,15) + d(0,2,5)$

(2*5), (CO1)

Q3. (a) Draw full adder circuit by using half adder.

(b) Draw the logic circuit design of Full subtractor.

(2*5), (CO2)

Q4. Define Fan-in, Fan-out, Propagation delay, Noise Margin, Figure of Merit for digital logic families.

(10), (CO2)

Q5. Explain working of BCD adder and Parallel adder.

(10), (CO2)

Course advisor: Rita Devi